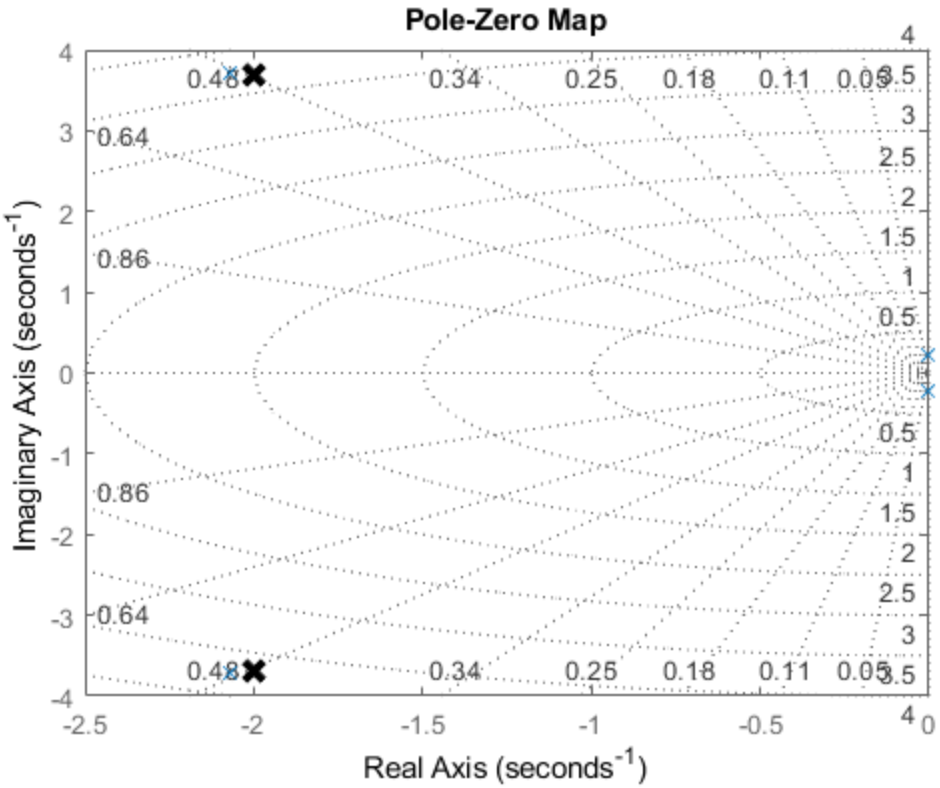
Introduction To Flight Mechanics

Tutorial 10



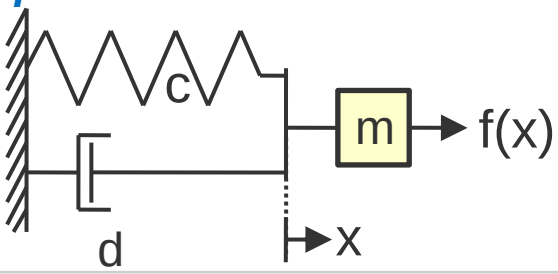
source: flugrevue.de

The following figure shows the pole pair of an eigenmode of the **longitudinal motion**:



a) Which eigenmode is it?

- ***Equation of Motion:***

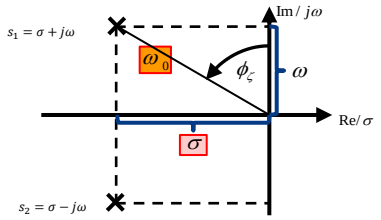


Spring constant	c
Damping coefficient	d
Mass	m
Deflection	x

$$m\ddot{x} + d\dot{x} + cx = 0$$
$$\ddot{x} + \frac{d}{m}\dot{x} + \frac{c}{m}x = 0$$

The differential equation of a linear 2nd order system can be written as:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = 0 \quad \text{or} \quad \ddot{x} - 2\sigma\dot{x} + \omega_0^2 x = 0$$



Comparing the coefficients leads to:

Natural Frequency: $\omega_0 = \sqrt{\frac{c}{m}}$ **Absolute Damping:** $\sigma = -\frac{1}{2}\frac{d}{m}$ **Relative Damping:** $\zeta = -\frac{\sigma}{\omega_0} = \frac{d}{2\cdot\sqrt{m\cdot c}}$

Characteristic Polynomial:

$$\lambda^2 + \frac{d}{m}\cdot\lambda + \frac{c}{m} = 0 \quad \Rightarrow \lambda^2 + 2\zeta\,\omega_0\cdot\lambda + \omega_0^2 = 0 \quad \Rightarrow \quad \lambda_{1,2} = \sigma \pm j\,\omega = -\zeta\,\omega_0 \pm j\,\omega_0\sqrt{1-\zeta^2} \quad \Rightarrow \quad \lambda_{1,2} = -\frac{d}{2m} \pm \sqrt{\left(\frac{d}{2m}\right)^2 - \frac{c}{m}}$$

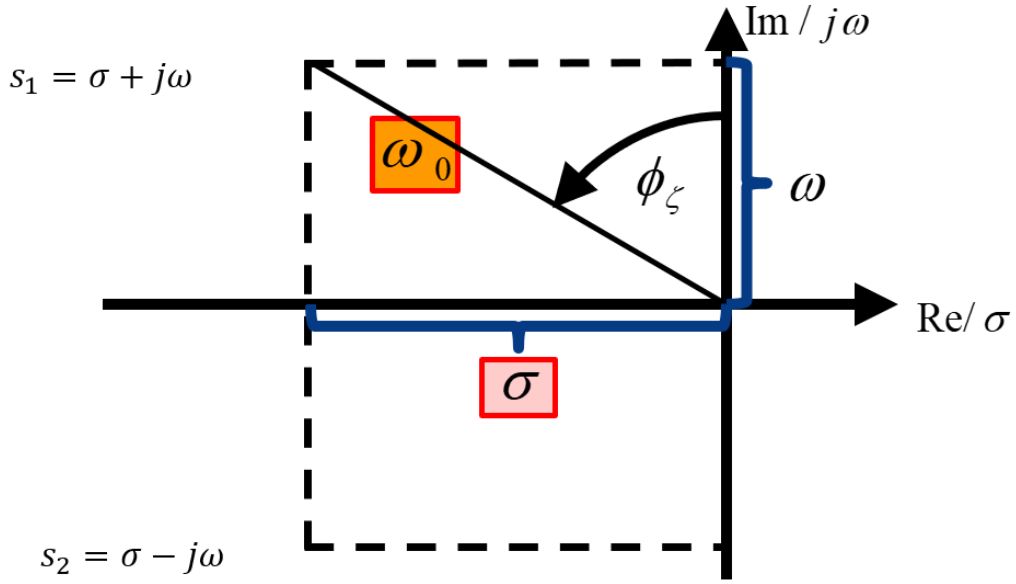
Eigenvalues

- ***State Space Model:***

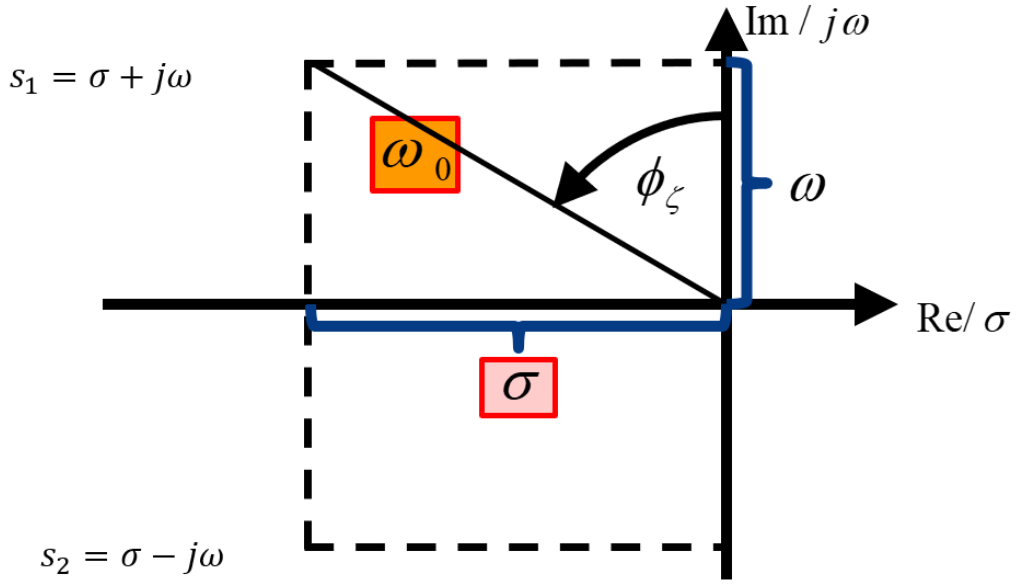
$$\begin{bmatrix}\dot{x}\\ \ddot{x}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{c}{m} & -\frac{d}{k}\end{bmatrix}\begin{bmatrix}x\\ \dot{x}\end{bmatrix}$$

Exercise 10.1

Calculate the characteristic polynomial for a system with a conjugate complex pole pair: $s_1 = \sigma + j\omega$ and $s_2 = \sigma - j\omega$



Calculate the characteristic polynomial for a system with a conjugate complex pole pair: $s_1 = \sigma + j\omega$ and $s_2 = \sigma - j\omega$



$$(s - s_1)(s - s_2) = (s - \sigma - j\omega)(s - \sigma + j\omega)$$

$$= s^2 - s\sigma + js\omega - \sigma s + \sigma^2 - \sigma j\omega - j\omega s + j\omega\sigma + \omega^2$$

$$= s^2 - s\sigma - \sigma s + \sigma^2 + \omega^2 = s^2 - 2\sigma s + \sigma^2 + \omega^2 = s^2 - 2\sigma s + \omega_0^2$$

$$= s^2 + 2\zeta\omega_0 s + \omega_0^2 \qquad \zeta = \frac{-\sigma}{\omega_0}$$

Exercise 10.1



What are the names of the two typical eigenmodes of the longitudinal motion of an aircraft?

Which states are contributing to which eigenmotion?

What are the characteristics of the eigenmodes? (are they oscillatory or aperiodic? Are they slow or fast? etc.)

$$\begin{bmatrix} \dot{V} \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} X_V & -g & X_\alpha & X_q \\ -Z_V & 0 & -Z_\alpha & -Z_q \\ Z_V & 0 & Z_\alpha & Z_q + 1 \\ M_V & 0 & M_\alpha & M_q \end{bmatrix} \cdot \begin{bmatrix} V \\ \gamma \\ \alpha \\ q \end{bmatrix}$$

Recap: Longitudinal Eigenmotion (II)

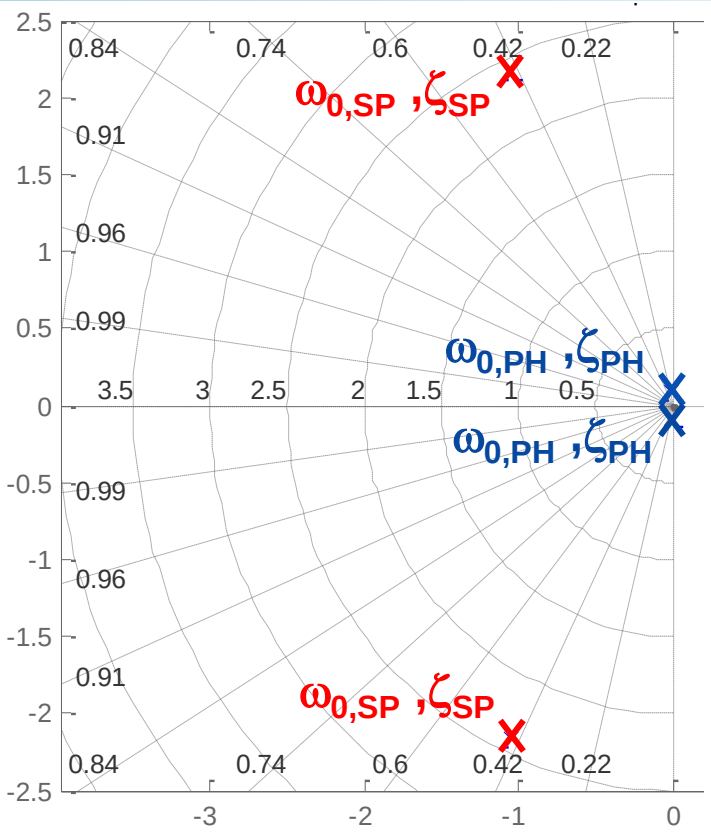
Typical position of the eigenvalues of a conventional aircraft in the complex plane:



source: www.airliners.net

Example Aircraft and Trim Data:

Aircraft:	DC 8
Mach Number:	0.443
Velocity:	142,6464 [m/s]
Mass:	86182,55 [kg]
Altitude:	4,57200 [km]



- **Two pairs of complex conjugate eigenvalues**
  → Two periodic and stable eigenmodes
- **The two different eigenmodes have highly different natural frequencies**

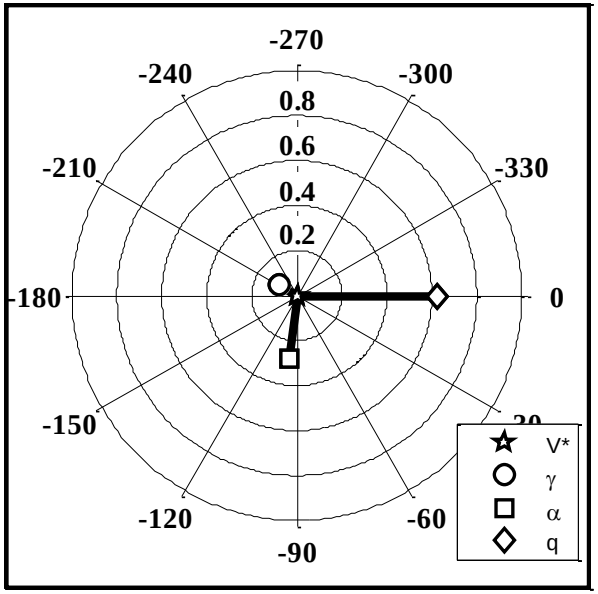
	Re(λ) [-]	Im(λ) [-]	ζ [-]	ω_0 [rad/s]
1	-0.0027447	0.089143	0.030776	0.089185
	-0.0027447	-0.089143	0.030776	0.089185
2	-1.043	2.1613	0.43462	2.3998
	-1.043	-2.1613	0.43462	2.3998

Slow periodic (low frequency), weakly damped
Phugoid (PH)

Fast periodic (high frequency), strongly damped
Short Period (SP)

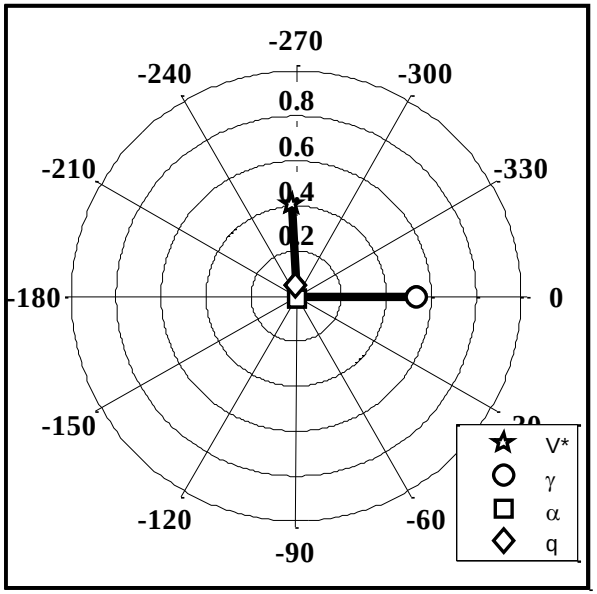
Recap: Longitudinal Eigenmotion (III)

Short Period



	|x|	arg(x) [°]
V*	0.0039838	-68.8305
γ	0.089972	-213.2013
α	0.28484	-97.5079
q	0.62121	0

Phugoid

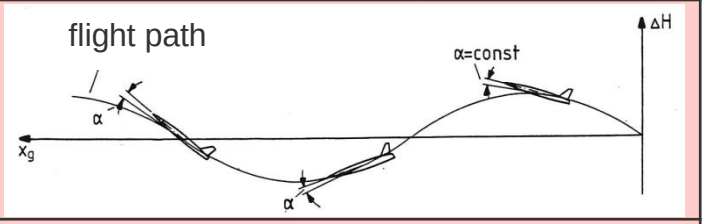


	|x|	arg(x) [°]
V*	0.41054	-267.7402
γ	0.53246	0
α	0.0094243	-84.4333
q	0.047576	-269.2439

Remark: Here we use the transformation $V^* = V/V_0$, with V_0 being the velocity at the trim point.

- **The main contributors to the short period are the angle of attack and the pitch rate**
- **The main contributors to the Phugoid are the velocity and the flight path angle**

Recap: Longitudinal Eigenmotion (V)

Eigenmodes of the Longitudinal Motion	**Short Period**	**Phugoid**
		
Damping	Strongly damped	Weakly damped
Time Period (frequency)	Small period time ($T_{DC\,8\ Short\ Period} \approx 3$ sec) (higher frequency)	High period time ($T_{DC\,8\ Phygoide} \approx 70$ sec) (small frequency)
Contribution of the velocity V	Small contribution (corresponding elements are small in the eigenvector)	High contribution (corresponding elements are high in the eigenvector)
Contribution of the flight path angle γ	Small contribution (corresponding elements are small in the eigenvector)	High contribution (corresponding elements are high in the eigenvector)
Contribution fo the angle of attack α	High contribution (corresponding elements are high in the eigenvector)	Small contribution (corresponding elements are small in the eigenvector)
Contribution of the pitch rate q	High contribution (corresponding elements are high in the eigenvector)	Small contribution (corresponding elements are small in the eigenvector)

Complementary Attributes of Short Period and Phugoid

Conclusion

Recap: Longitudinal Eigenmotion (VI)

	Short Period	Phugoid
	The period time of the short period is significantly smaller than the period time of the Phugoid, the contribution of the primary states of the Phugoid on the short period V and γ are very small	The period time of the short period is significantly smaller than the period time of the Phugoid, the contribution of the primary states of the Phugoid on the short period V and γ are very small The states α and q decay very fast when looking at the Phugoid time scale. Because of this, α and q can be neglected for a simplified model of the Phugoid.
Used states for the simplification	α and q	V and γ

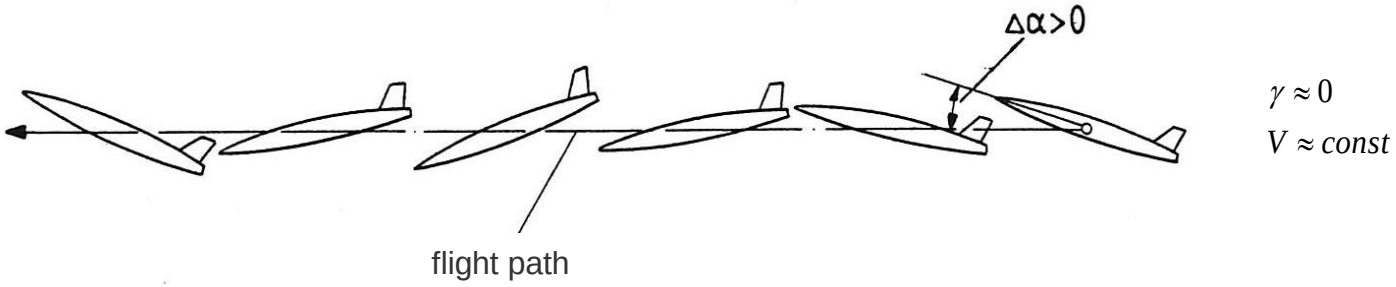
⇒ Simplification of the eigenmodes of the longitudinal motion via dividing the state space model

$$\begin{bmatrix}\dot{V}\\ \dot{\gamma}\\ \dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}X_V & -g & X_\alpha & X_q\\ -Z_V & 0 & -Z_\alpha & -Z_q\\ Z_V & 0 & Z_\alpha & Z_q+1\\ M_V & 0 & M_\alpha & M_q\end{bmatrix}\cdot\begin{bmatrix}V\\ \gamma\\ \alpha\\ q\end{bmatrix} + \begin{bmatrix}X_\eta & X_{\delta T}\\ -Z_\eta & -Z_{\delta T}\\ Z_\eta & Z_{\delta T}\\ M_\eta & 0\end{bmatrix}\cdot\begin{bmatrix}\eta\\ \delta_T\end{bmatrix}$$

Recap: Longitudinal Eigenmotion (IV)

Short Period:

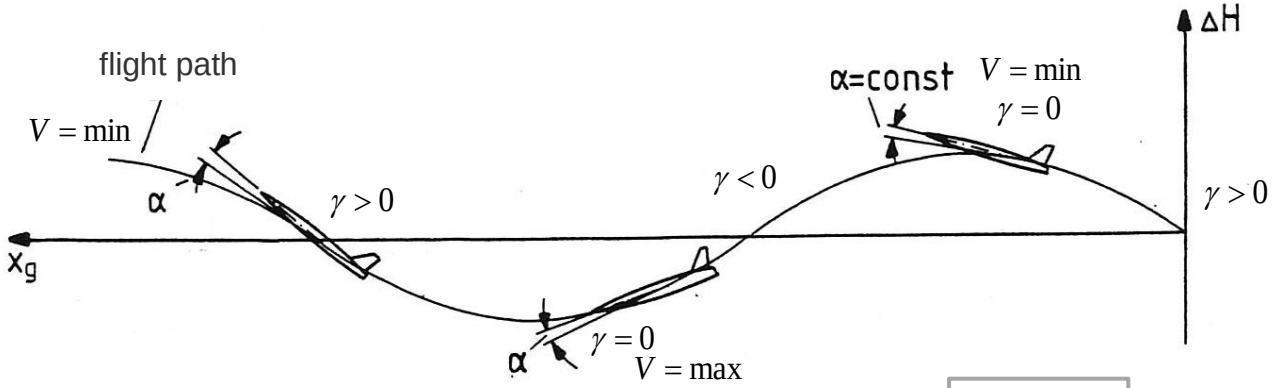
Simplified Visualization of the Short Period:



Period time short period of the reference configuration: T ≈ 3 sec

Phugoid:

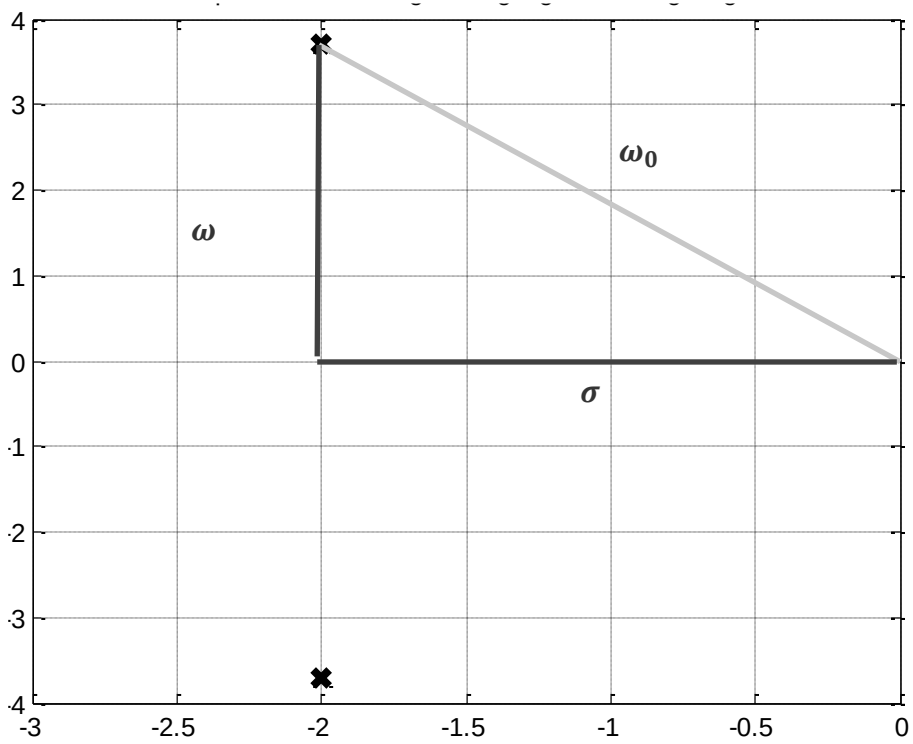
Simplified Visualization Phugoid:



Period time Phugoid of the reference configuration: T≈ 70 sec

a) Which eigenmode is it?

The damping and frequency properties are described by the eigenvalues: $\lambda = \sigma + i\omega$



From the plot we can read:

$$\sigma \approx -2$$

$$\omega \approx 3{,}7$$

$$\omega_0^2 = \sigma^2 + \omega^2$$

$$\omega_0 = \sqrt{\sigma^2 + \omega^2} = \sqrt{4 + 3{,}7^2}\,\frac{rad}{s} = 4{,}2059\,\frac{rad}{s}$$

$$T = \frac{2 \cdot \pi}{\omega_0} = \frac{2 \cdot \pi}{4{,}2059}\,s = 1{,}49\ s$$

„Order of magnitude“ for DC 8:

$$T_{\mathrm{PH,DC8}} \approx 61\ s$$
$$T_{\mathrm{SP,DC8}} \approx 3\ s$$

→ **Short Period**

b) Give an approximation for the derivatives, which define the eigenmode. You can neglect the influence of the pitch rate on the forces. Further simplifications have to be justified!

➔ eigenmode defined via ω_0 and σ

➔ by which derivatives are those parameters influenced?

b) Give an approximation for the derivatives, which define the eigenmode. You can **neglect the influence of the pitch rate on the forces**. Further simplifications have to be justified!

- From chapter 11 we know the reduced system:

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q + 1\\ M_\alpha & M_q\end{bmatrix}\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\eta$$

Hint: Know this by heart!

- We are also provided with the SP eigenmode!
  (from the previous exercise)

Compute the characteristic equation:

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}Z_\alpha & Z_q+1\\ M_\alpha & M_q\end{bmatrix}\begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}Z_\eta\\ M_\eta\end{bmatrix}\cdot\eta$$

$$(s\boldsymbol{I}-\boldsymbol{A}) = \begin{bmatrix}s-Z_\alpha & -(Z_q+1)\\ -M_\alpha & s-M_q\end{bmatrix}$$

$$\det(s\boldsymbol{I}-\boldsymbol{A}) = (s-Z_\alpha)\cdot(s-M_q) - (M_\alpha)\cdot(Z_q+1)$$

$$s^2 - M_q\cdot s - Z_\alpha\cdot s + M_q\cdot Z_\alpha - M_\alpha\cdot Z_q - M_\alpha = 0$$

The general characteristic equation of a 2nd order system :

$$\boxed{s^2 - 2\cdot\sigma\cdot s + \omega_0^2 = 0}$$

Comparison of the coefficients:

$$s^2 - \underbrace{(M_q + Z_\alpha)}_{2\cdot\sigma}\cdot s + \underbrace{M_q\cdot Z_\alpha - M_\alpha\cdot(Z_q+1)}_{\omega_{0,SP}^2} = 0$$

$$\omega_{0,SP}^2 = M_q\cdot Z_\alpha - M_\alpha\cdot(Z_q+1) \qquad 2\cdot\sigma = (M_q + Z_\alpha)$$

$$\omega_{0,SP}^2 = M_q \cdot Z_\alpha - M_\alpha \cdot (Z_q + 1)$$

$$2 \cdot \sigma = (M_q + Z_\alpha)$$

Eigenfrequency:

First simplification: $Z_q \approx 0$ (Given)

Second simplification : $M_q \cdot Z_\alpha$ is neglected as it is “smaller” than $M_\alpha \cdot 1$.
(“Justified” when comparing the magnitude of the two terms!)

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & \mathbf{-1.14} & 0.957 \\ 0.1 & 0 & \mathbf{-15.34} & \mathbf{-3.00} \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

$$\boldsymbol{M_{\alpha,exercise\ c)} = -15,34}$$

$$\boldsymbol{M_{q,exercise\ c)} \cdot Z_{\alpha,exercise\ c)} = -3.00 \cdot -1.14 = 3,42}$$

→ $\omega_{0,SP}^2 \approx -M_\alpha$

$M_\alpha \approx -\omega_{0,SP}^2 = -(4{,}2059^2) = -17{,}689$ (values from task a)

$$\omega_{0,SP}^2 = M_q \cdot Z_\alpha - M_\alpha \cdot (Z_q + 1)$$

$$2 \cdot \sigma = (M_q + Z_\alpha)$$

Damping:

Third (big) simplification: Z_α neglected → Error quite high

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & \mathbf{-1.14} & 0.957 \\ 0.1 & 0 & -15.34 & \mathbf{-3.00} \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

$$M_{q,Aufgabe\ c)} = \mathbf{-3,00}, \qquad Z_{\alpha,Aufgabe\ c)} = \mathbf{-1.14}$$

➡ Rough approximation of M_q: $2 \cdot \sigma \approx M_q \rightarrow M_q \approx 2 \cdot (-2) = -4$

(values from task a)

c) Does the pole zero plot in 10.1 belong to the following linear state space model of a Boeing 777?

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

Compute the eigenvalues of the B777 (using the reduced system)

$$\Rightarrow \quad \begin{bmatrix} \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} -1.14 & 0.957 \\ -15.34 & -3.00 \end{bmatrix} \cdot \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} 0.059 \\ 6.56 \end{bmatrix} \cdot \eta$$

Note: The “order” of the relevant entries depends on state vector order!

The derivatives are therefore:

$$\begin{bmatrix}\dot{\alpha}\\ \dot{q}\end{bmatrix} = \begin{bmatrix}-1.14 & 0.957\\ -15.34 & -3.00\end{bmatrix} \cdot \begin{bmatrix}\alpha\\ q\end{bmatrix} + \begin{bmatrix}0.059\\ 6.56\end{bmatrix} \cdot \eta$$

$$M_\alpha = -15{,}34$$

$$M_q = -3{,}00$$

$$Z_\alpha = -1{,}14$$

$$Z_q + 1 = 0{,}957$$

relations from exercise b)

$$\sigma_{SP} = \frac{1}{2}\left(M_q + Z_\alpha\right) = \frac{1}{2}(-3{,}00 - 1{,}14) = -2{,}07$$

$$\omega_{0,SP} = \sqrt{\left(M_q \cdot Z_\alpha - M_\alpha \cdot \left(Z_q + 1\right)\right)} = \sqrt{(3{,}00 \cdot 1{,}14 + 15{,}34 \cdot 0{,}957)} = 4.2545\,\frac{rad}{s}$$

(-2 and 4,2)

Compare with a) → ~same values → model appears to be the same

Exercise 10.3

Error due to simplification:

Name	Value of complete model	Value using the approximation in 10.2/ b)	Error in %
M_α	$-15{,}34$	$-17{,}689$	15,3 %
M_q	$-3{,}00$	-4	33,33 %
Z_α	$-1{,}14$	0	100 %
$Z_q + 1$	$0{,}957$	1	4,5 %

What we did: knowing only the eigenvalue location we calculated in a) ω_0 and σ. Then using the shortperiod approximation and assuming $Z_\alpha = Z_q = 0$ we calculated in b) M_α and M_q from ω_0 and σ. In c) the true full statespace model was given and we could calculate the true ω_0 and σ.

- **State space model of the lateral motion from linearising the nonlinear equations of motion**

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} N_r & N_\beta & N_p & 0 \\ -1 & Y_\beta & 0 & \frac{g}{V_0} \\ L_r & L_\beta & L_p & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} N_\xi & N_\zeta \\ Y_\xi & Y_\zeta \\ L_\xi & L_\zeta \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

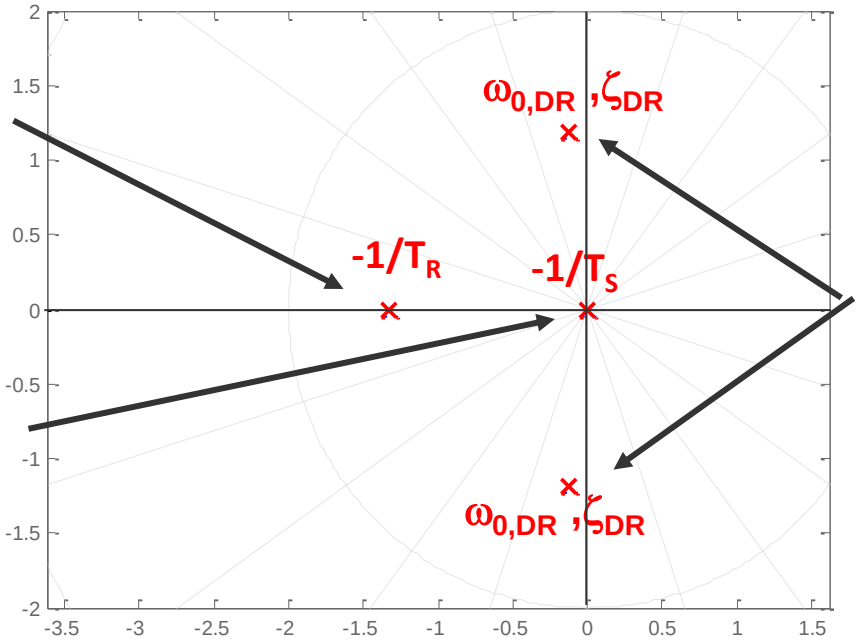
- **Characteristic equation / eigenvalues**

$$\det(sI - A) = 0 \Rightarrow \det \begin{bmatrix} s - N_r & -N_\beta & -N_p & 0 \\ 1 & s - Y_\beta & 0 & -\frac{g}{V_0} \\ -L_r & -L_\beta & s - L_p & 0 \\ 0 & 0 & -1 & s \end{bmatrix} = 0$$

Typical position of the

real eigenvalue $-1/T_R$: stable, aperiodic eigenmode

real eigenvalue $-1/T_S$: stable or instable, aperiodic eigenmode



complex conjugate eigenvalue pair $\omega_{0,DR}$,ζ_{DR}: stable, periodic eigenmode

	Re(λ) [-]	Im(λ) [-]	ζ [-]	ω_0 [rad/s]		
1	-1.3301	0	1	1.3301	fast, aperiodic	Roll Mode
2	-0.12742	-1.1906	0.10641	1.1974	fast, periodic	Dutch-Roll
	-0.12742	1.1906	0.10641	1.1974		
3	-0.0065518	0	1	0.0065518	slow, aperiodic	Spiral Mode

- **Characteristic Equation:**

  Based on the results of the last slide, the characteristic equation of the lateral motion can be given now. There are two real poles (poles on the real axis), which correspond to the Roll Mode and the Spiral Mode. Additionally there is a complex conjugate pole pair which corresponds to the Dutch-Roll.

$$\left(s+\frac{1}{T_R}\right)\left(s+\frac{1}{T_S}\right)\left(s^2+2\zeta_{DR}\omega_{0,DR}s+\omega_{0,DR}^2\right)=0$$

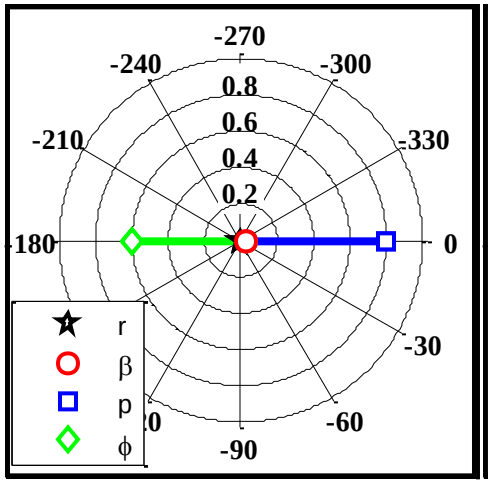
Pole of the Roll Mode

Pole of the Spiral Mode

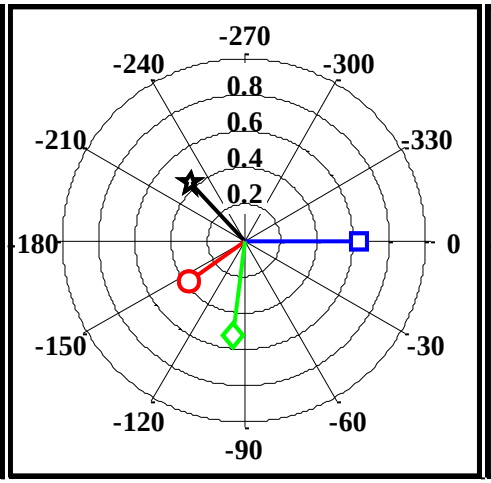
Pole of the Dutch-Roll

Recap: Lateral Eigenmotion (IV)

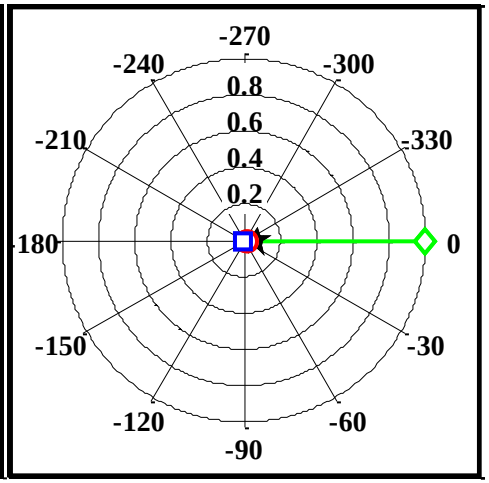
Roll Mode



Dutch-Roll



Spiral Mode



	|x|	arg(x) [°]
r	0.0075949	-180
β	0.027417	0
p	0.79897	0
ϕ	0.60069	-180

	|x|	arg(x) [°]
r	0.4256	-227.5901
β	0.37822	-142.3237
p	0.63098	0
ϕ	0.52696	-96.1085

	|x|	arg(x) [°]
r	0.067377	0
β	0.012808	0
p	0.0065362	-180
ϕ	0.99762	0

Exercise 10.3

The following linear model of the lateral motion is given:

$$\begin{pmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{pmatrix} = \begin{bmatrix} -0.6 & 6.0 & -0.18 & 0 \\ -1 & -0.19 & 0 & 0.13 \\ 1.35 & -15.10 & -6.5 & 0 \\ 0.05 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{pmatrix} r \\ \beta \\ p \\ \Phi \end{pmatrix} + \begin{bmatrix} 0.0075 & -2.6 \\ 0 & 0.037 \\ -13.09 & 1.87 \\ 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} \xi \\ \zeta \end{pmatrix}$$

Calculate the **initial roll acceleration** and the **steady state roll rate** for an aileron deflection of 1°.

Recap: The Roll Mode (I)

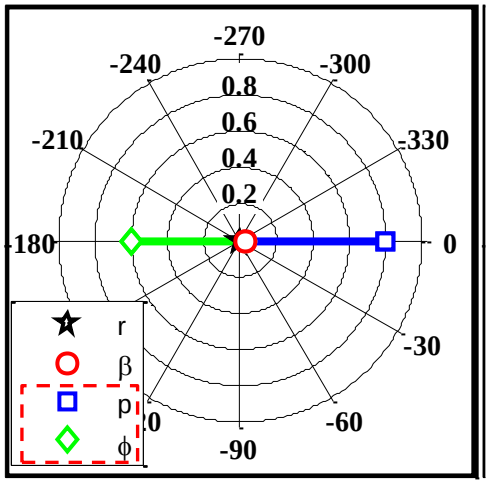
- **State space model lateral motion**

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} N_r & N_\beta & N_p & 0 \\ -1 & Y_\beta & 0 & \frac{g}{V_0} \\ L_r & L_\beta & L_p & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} N_\xi & N_\zeta \\ Y_\xi & Y_\zeta \\ L_\xi & L_\zeta \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

- In a first approximation, the main contributors in the Roll Mode are the **roll rate** p and the **roll angle** Φ (see eigenvectorstar of the Roll Mode).

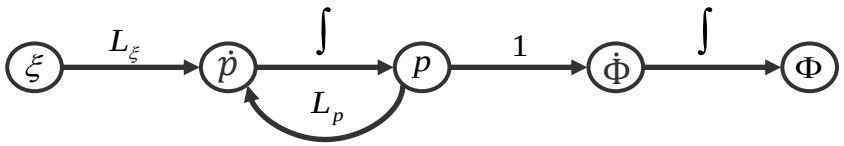
- The Roll Mode is mainly excited by an aileron deflection ξ.

Roll Mode

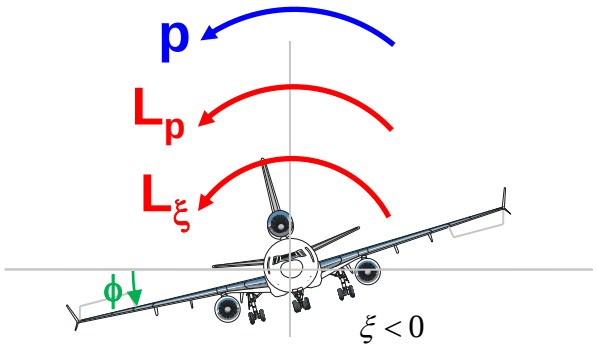


State space model for approximation of the roll mode

$$\begin{bmatrix} \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} L_p & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} p \\ \Phi \end{bmatrix} + \begin{bmatrix} L_\xi \\ 0 \end{bmatrix} \cdot \xi$$



States and moments of the aircraft



Characteristic equation

$$\det(s \cdot \boldsymbol{I} - \boldsymbol{A}) = \det \begin{bmatrix} s - L_p & 0 \\ -1 & s \end{bmatrix} = (s - L_p) \cdot s$$

⇨ **Eigenvalue of the Roll Mode** $\lambda_R = L_P$ ⟶

⇨ **Rough approximation of the eigenvalue of the Spiral Mode (S) as integrator pole**

$\lambda_S = 0$

Roll Time Constant

$$T_R = -\frac{1}{L_P}$$

Typical Values: 0,2 ÷1,5 [s]

Recap: The Roll Mode (III)

State space model of the approximation of the Roll Mode

$$\begin{bmatrix}\dot{p}\\ \dot{\Phi}\end{bmatrix} = \begin{bmatrix}L_p & 0\\ 1 & 0\end{bmatrix}\cdot\begin{bmatrix}p\\ \Phi\end{bmatrix} + \begin{bmatrix}L_\xi\\ 0\end{bmatrix}\cdot\xi$$

$$\frac{p(s)}{\xi(s)} = \frac{L_\xi}{s - L_p} = -\frac{L_\xi}{L_p}\cdot\frac{1}{T_R\cdot s + 1}$$

Roll moment equation

$$\dot{p} = L_P\cdot p + L_\xi\cdot\xi$$

L

Laplace Transformation

$$s\cdot p(s) = L_p\cdot p(s) + L_\xi\cdot\xi(s)$$
$$p(s)\cdot(s - L_p) = L_\xi\cdot\xi(s)$$

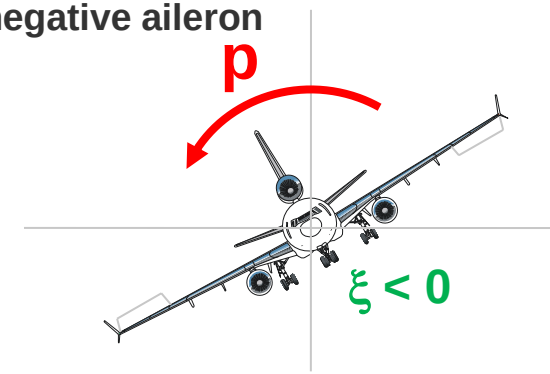
Transfer function from aileron ξ to the roll rate p

$$\frac{p(s)}{\xi(s)} = \frac{L_\xi}{s - L_p} = -\frac{L_\xi}{L_p}\cdot\frac{1}{T_R s + 1}$$

PT_1 – behavior

Delay of first order as approximation of Roll Mode

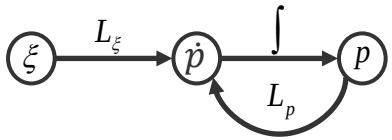
Aircraft reaction to a negative aileron deflection (ξ<0):

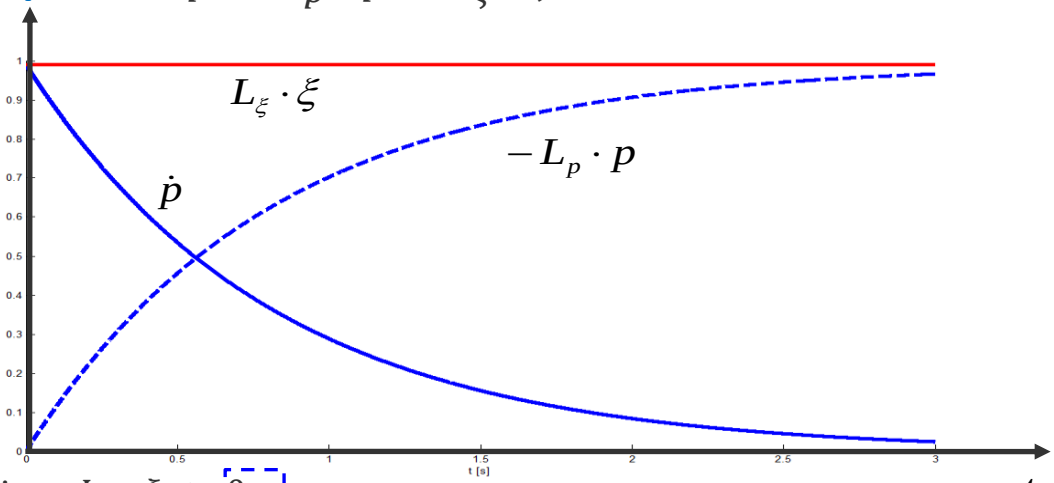


A negative aileron deflection (ξ<0) leads to a positive roll rate p

Negative aileron deflection (ξ<0) means:
- negative surface deflection on the right side
- positive surface deflection on the left side

Explanation of the roll moment equation: $\dot{p} = L_p \cdot p + L_\xi \cdot \xi$





Negative aileron deflection $\xi_0 < 0$

< 0

➔ Positive roll moment: $L_\xi \cdot \xi_0 > 0$

➔ Positive initial roll acceleration: $\dot{p}_0 = L_\xi \cdot \xi_o > 0$

< 0

➔ Build up of a positive roll rate p: $p = \int_0^t \dot{p} dt > 0$

➔ Positive roll rate leads to a negative roll damping: $L_p \cdot p < 0$

➔ Increasing roll rate p leads to a constant increase of roll damping moment until the moment induced by roll damping equals the moment produced by the aileron deflection leading to a zero roll acceleration.

$$L_p \cdot p = -L_\xi \cdot \xi$$
$$\Rightarrow \dot{p}_\infty = 0$$

➔ **Stationary roll rate p_∞**

Step response of the roll rate

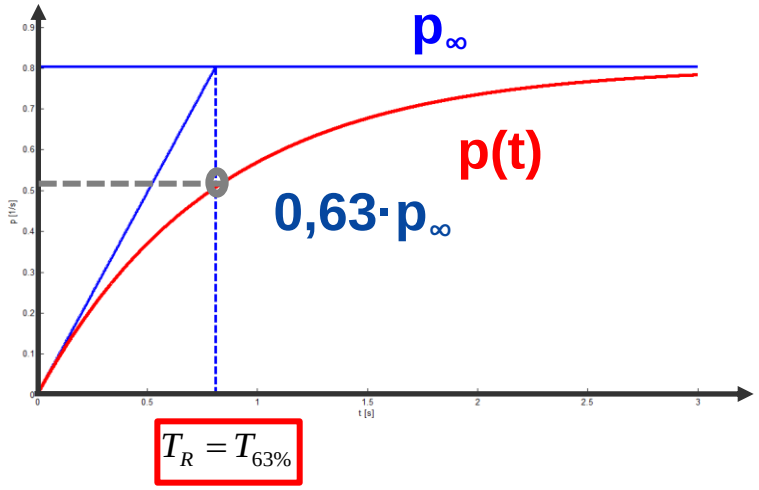
Aileron effectiveness

$$p(t) = -\frac{L_\xi}{L_p} \cdot \xi_0 \cdot \left(1 - e^{-\frac{t}{T_R}}\right)$$

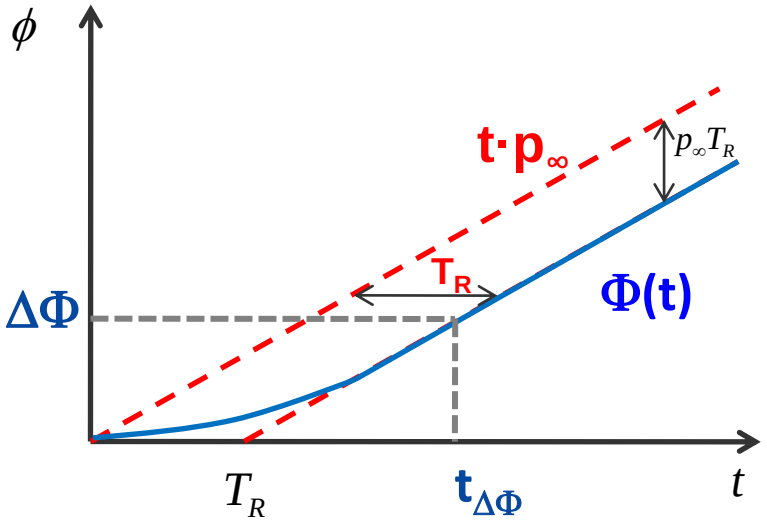
Roll damping

$$T_R = -\frac{1}{L_p}$$

Roll time constant from simplification of Roll Mode



- The roll time constant is the point in time, at which the roll rate equals 63% of the steady state roll rate (due to a step input of the aileron)
- The higher the roll time constant T_R, the longer the aircraft takes to achieve a rollrate after an aileron command
- The bank angle follows from integrating the roll rate p



$$\dot{p} = L_p \cdot p + L_\xi \cdot \xi$$

given the aileron deflection $\bar{\xi} = 1°$ we can compute:

the **initial roll acceleration** $\dot{p}_0$ → here, the roll rate is $p = 0$:

$$\dot{p}_0 = L_\xi \cdot \bar{\xi} = -13.09 \frac{1}{s^2} \cdot 1° = -13.09 \ °/s^2$$

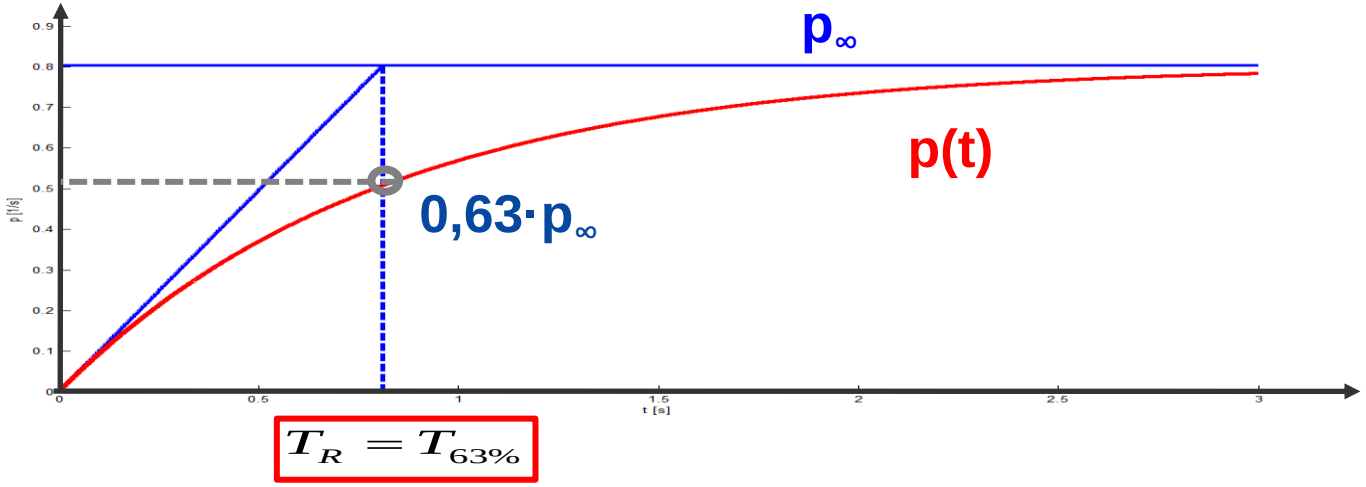
the **steady-state roll rate** p_∞ → here, the roll acceleration is $\dot{p} = 0$:

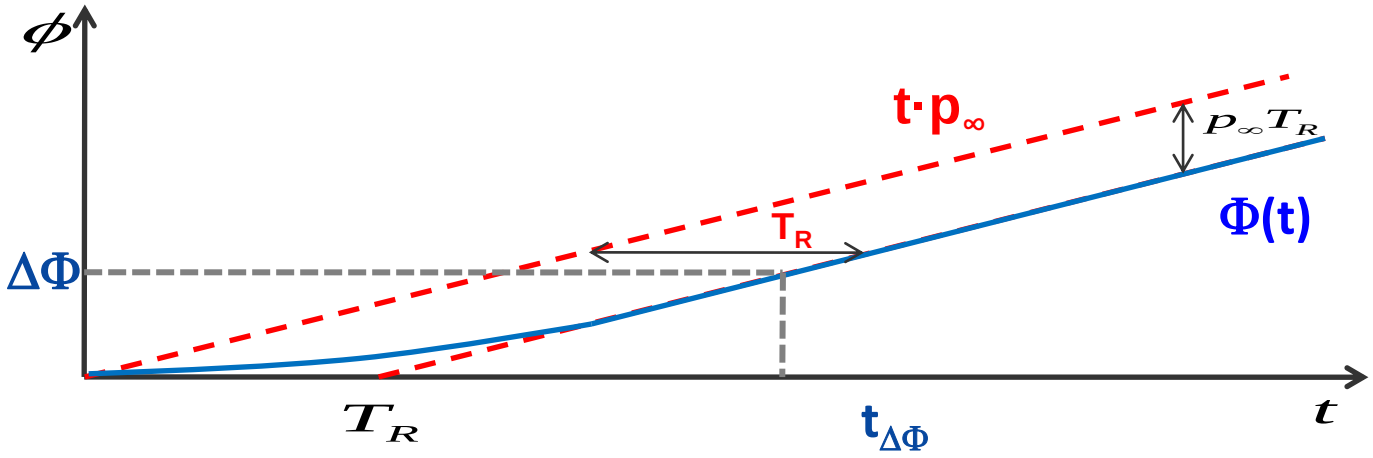
$$0 = L_p \cdot p_\infty + L_\xi \cdot \bar{\xi}$$

$$p_\infty = -\frac{L_\xi}{L_p} \cdot \bar{\xi}$$

$$p_\infty = -\frac{-13.09}{-6.5} \cdot 1° = -2.0138 \ °/s$$

Plot the step response of roll rate and bank angle for the given aileron deflection.

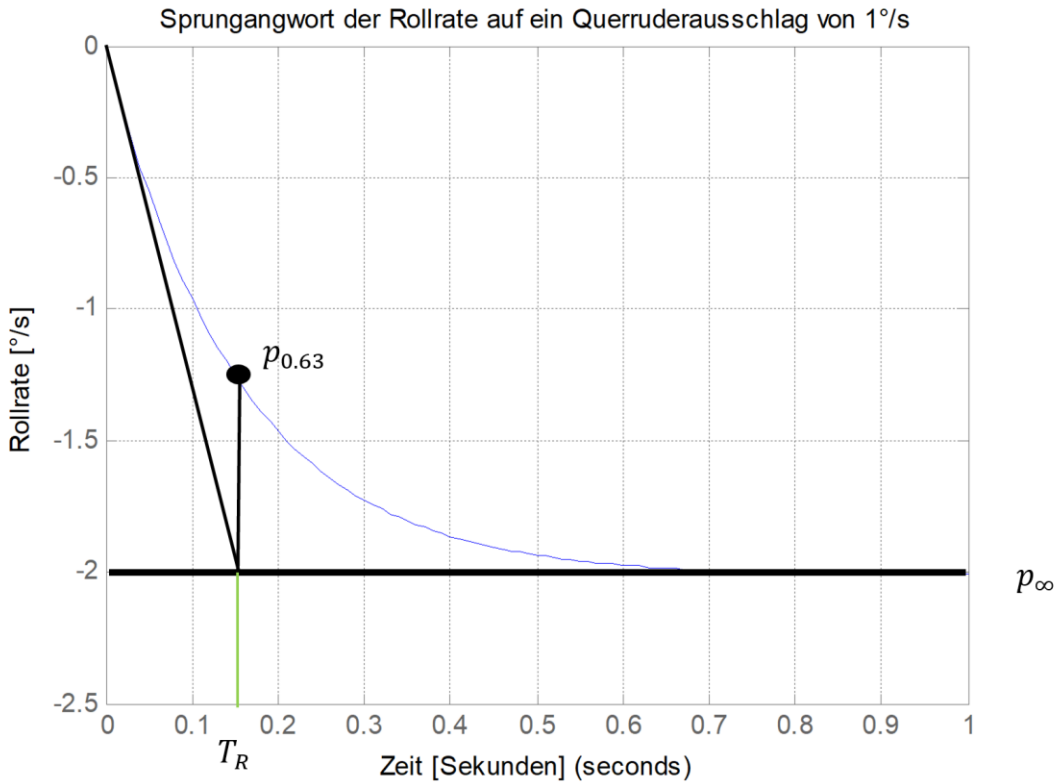




Exercise 10.4

1) Roll time constant : $T_R = -\frac{1}{L_P} = -\frac{1}{-6,5} = 0.1538$

2) Roll rate at T_R: $p_{TR} = 0,63 \cdot p_\infty = 0,63 \cdot -2.0138 = 1,2687 \frac{°}{s}$



Exercise 10.4

Time for p = p_∞ → from figure last slide: approx 0.7 s

